RAJALAKSHMI ENGINEERING COLLEGE

Nandha Kishore S 251001157-2025-IT

N2

| | |
|---|---|
| **Started on** | Friday, 11 September 2026, 3:36 PM |
| **State** | Finished |
| **Completed on** | Friday, 11 September 2026, 4:02 PM |
| **Time taken** | 25 mins 14 secs |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 10.00 out of 10.00

**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

**Example 1:**

***Input:*** *6*

***Output:****6*

***Explanation:*** *There are 6 ways to 6 represent number with 1 and 3*

*1+1+1+1+1+1*

*3+3*

*1+1+1+3*

*1+1+3+1*

*1+3+1+1*

*3+1+1+1*

**Input Format**

First Line contains the number n

**Output Format**

**Print: The number of possible ways 'n' can be represented using 1 and 3**

Sample Input

6

Sample Output

6

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>


long long count_ways(int n) {
    if (n < 0) return 0;
    if (n == 0 || n == 1 || n == 2) return 1;
    if (n == 3) return 2;


    long long dp[n + 1];


    dp[0] = 1;
    dp[1] = 1;
    dp[2] = 1;
    dp[3] = 2;

    for (int i = 4; i <= n; i++) {
        dp[i] = dp[i - 1] + dp[i - 3];
    }

    return dp[n];
}

int main() {
    int n;
```

```
    int n;


    if (scanf("%d", &n) == 1) {

        printf("%lld\n", count_ways(n));
    }

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 6 | 6 | 6 | ✔ |
| ✔ | 25 | 8641 | 8641 | ✔ |
| ✔ | 100 | 24382819596721629 | 24382819596721629 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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|---|---|
| **Started on** | Friday, 11 September 2026, 4:03 PM |
| **State** | Finished |
| **Completed on** | Friday, 11 September 2026, 4:06 PM |
| **Time taken** | 3 mins 27 secs |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 10.00 out of 10.00

**Playing with Chessboard:**

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

**Example:**

**Input**

3

**1** 2 4

**2** 3 4

**8 7 1**

**Output:**

19

**Explanation:**

Totally there will be 6 paths among that the optimal is

Optimal path value:1+2+8+7+1=19

**Input Format**

First Line contains the integer n

The next n lines contain the n*n chessboard values

**Output Format**

Print Maximum monetary value of the path

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>


int max(int a, int b) {
    return (a > b) ? a : b;
}

int main() {
    int n;

    if (scanf("%d", &n) != 1) return 0;


    int dp[n];

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            int current_value;
            scanf("%d", &current_value);

            if (i == 0 && j == 0) {

                dp[j] = current_value;
            } else if (i == 0) {

                dp[j] = dp[j-1] + current_value;
            } else if (j == 0) {

                dp[i] = dp[i] + current_value;
```

```
            } else {

                dp[j] = current_value + max(dp[j], dp[j-1]);
            }
        }
    }

    printf("%d\n", dp[n-1]);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 2 4<br>2 3 4<br>8 7 1 | 19 | 19 | ✔ |
| ✔ | 3<br>1 3 1<br>1 5 1<br>4 2 1 | 12 | 12 | ✔ |
| ✔ | 4<br>1 1 3 4<br>1 5 7 8<br>2 3 4 6<br>1 6 9 0 | 28 | 28 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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| | |
|---|---|
| **Started on** | Friday, 25 September 2026, 4:12 PM |
| **State** | Finished |
| **Completed on** | Friday, 2 October 2026, 4:04 PM |
| **Time taken** | 6 days 23 hours |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| s1 | | a | g | **g** | **t** | **a** | **b** |
| s2 | **g** | x | **t** | **x** | **a** | **y** | **b** |

**The length is 4**

Solveing it using Dynamic Programming

**For example:**

| Input | Result |
|---|---|
| aab<br>azb | 2 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>

#define MAX 1001

int max(int a, int b) {
    return (a > b) ? a : b;
}

int main() {
    char s1[MAX], s2[MAX];
    scanf("%s", s1);
    scanf("%s", s2);

    int m = strlen(s1);
    int n = strlen(s2);
    int dp[MAX][MAX];

    for (int i = 0; i <= m; i++) {
        for (int j = 0; j <= n; j++) {
            if (i == 0 || j == 0)
                dp[i][j] = 0;
            else if (s1[i - 1] == s2[j - 1])
                dp[i][j] = dp[i - 1][j - 1] + 1;
            else
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
        }
    }

    printf("%d\n", dp[m][n]);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | aab<br>azb | 2 | 2 | ✔ |
| ✔ | ABCD<br>ABCD | 4 | 4 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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N2

| | |
|---|---|
| **Started on** | Friday, 2 October 2026, 4:05 PM |
| **State** | Finished |
| **Completed on** | Friday, 2 October 2026, 4:14 PM |
| **Time taken** | 9 mins 13 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n], dp[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    int best = 0;
    for (int i = 0; i < n; i++) {
        dp[i] = 1;
        for (int j = 0; j < i; j++) {
            if (a[j] <= a[i] && dp[j] + 1 > dp[i])
                dp[i] = dp[j] + 1;
        }
        if (dp[i] > best)
            best = dp[i];
    }

    printf("%d\n", best);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 9<br>-1 3 4 5 2 2 2 2 3 | 6 | 6 | ✔ |
| ✔ | 7<br>1 2 2 4 5 7 6 | 6 | 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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